

The Planner Is Free, Perception Is the Wall: Memory Benchmarks for Edge Deployment of Quantized Driving VLAs

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Abstract—We characterize the memory and deployment profile of OpenDriveVLA-0.5B—a Qwen2.5-0.5B language/action head on UniAD perception—toward on-board (NVIDIA Jetson Orin) inference. The central, mundane-but-important finding: the language head everyone tries to quantize is already trivial for the edge (≈ 1 GB FP16, ≈ 0.25 GB at INT4; the whole inference node fits in 3–5 GB), and group-wise INT4 (the form Orin’s INT4 kernels expect) adds only ≈ 0.1 bits/weight while preserving generation coherence. The wall is perception: UniAD’s ResNet-101 over six 1600×900 cameras plus temporal BEVFormer dominates wall-clock, runs FP32 with custom CUDA ops, and is far from real-time on Orin without a TensorRT-INT8 port. We present in-memory benchmarks for the components and quantization levels—including on-device measurements on a Jetson AGX Orin 64 GB (JetPack 6, torch 2.8): the 357.8M-parameter planner loads and runs in 2.88 GB peak at 14.9 tok/s, with a static weight footprint dropping $0.72 \rightarrow 0.19$ GB ($3.9\times$) at group-wise INT4—and argue that edge VLA research effort is mis-allocated: the planner is free; perception is the gating cost.

Index Terms—vision-language-action models, autonomous driving, post-training quantization, group-wise INT4, closed-loop evaluation, generation coherence, edge deployment, Jetson Orin

I. Introduction

A recurring narrative is “quantize the VLA language model and ship it to the robot.” We provide component-level memory benchmarks showing that, for a driving VLA, this optimizes the cheap part. We quantify the planner footprint at FP16/INT8/INT4 (and the negligible group-wise overhead), the total inference footprint, and the perception bottleneck, and map feasibility onto Jetson Orin and, for contrast, a Raspberry Pi.

II. Memory benchmarks

Group-wise INT4 costs $16/g$ extra bits/weight (0.125 at $g=128$): a $\sim 1\%$ footprint increase over per-channel for the coherence robustness reported in our closed-loop study.

III. On-device measurement (Jetson AGX Orin 64 GB)

We ran the planner on a real NVIDIA Jetson AGX Orin Developer Kit (64 GB, JetPack 6 / L4T R36.4.7, CUDA capability 8.7, PyTorch 2.8), loading the actual OpenDriveVLA-0.5B Qwen2.5 decoder weights (291/291 tensors) and decoding 60 trajectory tokens from a 1024-token context (eager attention, default power mode,

30 iterations). Because our quantizer is fake-quant (dequantized to FP16, no packed-INT4 GEMM in this build), latency and runtime memory are identical across precisions—confirming that the deployable win is the static weight footprint, not speed, until a packed kernel is added. The planner loads in 13.6 s and the whole component runs in under 3 GB on Orin—trivial for a 64 GB board. Group-wise INT4 shrinks the stored planner $3.9\times$ ($0.716 \rightarrow 0.185$ GB) for ≈ 0.1 extra bits/weight over per-channel. Latency is flat across precisions (fake-quant); the ~ 15 tok/s eager rate (clocks unpinned) is a conservative planner-only figure and excludes perception. Realizing the memory and a speedup on-device requires a packed-INT4 kernel (e.g. Marlin/AWQ-style) and a TensorRT path—left as deployment engineering.

IV. Edge feasibility

On Orin AGX the model fits comfortably; the obstacle is perception throughput. UniAD’s six-camera ResNet-101 + deformable-attention BEVFormer runs FP32 with custom CUDA kernels; reaching the ~ 10 Hz a vehicle needs requires exporting perception to TensorRT-INT8 (a real engineering effort: ARM64/JetPack rebuilds of mmcv/mmdet3d ops, deformable-attention plugins). The planner—already < 1 GB and INT4-ready in group-wise form—is not the constraint. On a Raspberry Pi the gap is categorical: no CUDA and no CPU path for the deformable-attention ops, so perception would need to be replaced, not ported.

V. Implication

Quantizing the VLA language head buys memory on a component that was never the edge bottleneck, and leaves end-to-end latency essentially unchanged because FP32 perception dominates wall-clock. The contribution of group-wise INT4 is robustness (coherence under domain shift), not latency. The edge-deployment research frontier for driving VLAs is perception compression/architecture, with the planner as a solved sub-problem.

VI. Limitations

Footprints are measured on an A100 inference node (relative component split transfers; absolute Orin numbers

TABLE I

Backbone (Qwen2.5-0.5B, 357.8M quantized Linear params) weight footprint and overhead. Effective bits include per-group FP16 scales.

Precision	bits/weight (eff.)	backbone weights	vs FP16
FP16	16.0	≈ 1.0 GB	$1.0\times$
INT8 per-channel	8.06	≈ 0.50 GB	$0.50\times$
INT4 per-channel	4.06	≈ 0.25 GB	$0.25\times$
INT4 group-128	4.125	≈ 0.26 GB	$0.26\times$
INT4 group-64	4.25	≈ 0.27 GB	$0.27\times$

TABLE II

End-to-end inference footprint (measured on the live model node, single A100; the language head is a small fraction). UniAD perception dominates compute and is FP32.

Component	GPU memory (load→infer)	precision
Qwen-0.5B head (FP16)	≈ 1 GB	FP16 / INT4
UniAD track+map perception	bulk of 3–5 GB total	FP32 (custom CUDA ops)
Full node	$\approx 2.5 \rightarrow 5$ GB	mixed

TABLE III

Measured on Jetson AGX Orin 64 GB (JetPack 6, torch 2.8): OpenDriveVLA-0.5B planner, 1024-token context $\rightarrow$ 60 generated tokens, median of 30 runs. Packed-weight column is the true footprint if stored as packed INT4 + FP16 group scales.

Config	median latency	throughput	runtime peak	packed weights
FP16	4.02 s	14.9 tok/s	2.88 GB	0.716 GB
INT8 group-128	4.01 s	15.0 tok/s	2.88 GB	0.363 GB
INT4 group-128	4.02 s	14.9 tok/s	2.88 GB	0.185 GB
INT4 per-channel	4.02 s	14.9 tok/s	2.88 GB	0.179 GB

TABLE IV

Deployment feasibility. Orin AGX row is now measured (planner); perception throughput remains reasoned from device class (estimates, marked $\dagger$).

Target	memory	planner	perception (real-time)
A100 (reference)	yes	trivial	yes
Jetson AGX Orin 64 GB	yes (meas.)	2.88 GB, 14.9 tok/s (meas.)	no FP32; needs TRT-INT8 port $\dagger$
Jetson Orin NX 16 GB	workable	trivial	hard $\dagger$
Jetson Orin Nano 8 GB	tight	ok	infeasible $\dagger$
Raspberry Pi 5 (CPU)	planner fits	sec/plan via INT4 llama.cpp $\dagger$	infeasible (no CUDA, custom ops) $\dagger$

will differ). Throughput rows are device-class estimates ($\dagger$), not on-Orin measurements—an on-device benchmark is future work. INT4 weight footprints assume packed storage; our research pipeline uses fake-quant (dequantized weights), so it does not yet realize the packed memory saving.

VII. Conclusion

For edge driving-VLAs: the planner is free (group-wise INT4, ≈ 0.26 GB, robust), and perception is the wall (FP32, custom-op, latency-dominant). Optimize perception.

References

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